

WHICH TOILETS DELIVER?

A new testing procedure gives researchers a valuable tool to evaluate the effectiveness of low-flow toilets.



Water

BY BILL GAULEY

When builders install new toilets in new construction or in existing homes, they need to be able to trust the stated efficiency and effectiveness of those toilets, since their reputations and businesses depend on the trust of their customers. Conventional 3.5-gallon (13-liter) toilets consume approximately 33% of all household water. More water-efficient toilets can reduce that figure to 18%. So, in an effort to conserve water, many consumers are choosing water-efficient 1.6-gallon (6-liter) ultralow-flush (ULF) toilets when building or remodeling their homes. In fact, many municipalities across North America now offer financial rebates for ULF retrofits, and the United States, the Province of Ontario, and some cities in Canada mandate such toilets in all new construction. But are those toilets actually delivering the promised water savings?

Virtually all the toilet models sold in North America meet both the flush vol-

ume and the performance requirements of the Canadian Standards Association (CSA) and the American National Standards Institute/American Society of Mechanical Engineers (ANSI/ASME). However, recent research in Canada and the United States has shown that some certified and commercially available models do not flush effectively, leading to customer complaints and the need for double-flushing (see “Double Flushing Dilemma,” *HE* Sept/Oct '02, p. 10). Unfortunately, the water savings promised to consumers and water agencies by many low-flush toilet manufacturers are going down the drain.

Currently, there is no convenient way for the customer to distinguish between good and marginal performers when buying a toilet. This lack of information on toilet performance has served to create a general negative impression of low-flush technology in the public mind, when in fairness, only the lackadaisical performers should rightfully be singled out.

Another toilet performance issue is the effect of flapper replacement on flush performance. Flappers wear out and need to be replaced approximately every five years. Toilets that use adjustable flappers may lose water efficiency if the adjustable flappers are replaced with a standard flapper—the type typically found at most home supply stores.

In 2003, the Canada Mortgage and Housing Corporation (CMHC) organized a consortium of 22 water agencies from North America to address the problem of poor performance by testing the effectiveness of low-flush toilets. The result, the Maximum Performance Testing of Popular Toilet Models (MaP) program, was led by the Canadian Water and Wastewater Association (CWWA) and sought to rank toilet makes and models based on tested flush performance. The results of testing by Veritec Consulting Incorporated, where I am a principal, provide much-needed information to consumers and builders about which low-flush toilets really work well.

Testing for the Real World

Previous toilet performance testing used unrealistic test media, such as sponges, plastic balls and beads, and Kraft paper. MaP uses soybean paste that has physical properties (density, moisture content) similar to those of human waste, in combination with toilet paper. Most people in the toilet industry agree that these media more accurately replicate real-world demands upon a toilet fixture.

In addition to using realistic test media, MaP adjusted all toilet samples rated at 1.6 gallons (6 liters) to flush at that volume before testing. This was to ensure a level playing field.

The ability of a toilet to completely remove waste in a single flush without plugging or clogging is generally considered one of the most important test criteria (there is currently no suitable test medium to simulate streaking). MaP identified 250 grams as the performance threshold. This value is based on the results of a British medical study that identified 250 g as the average maximum fecal size of the male participants in the study. Any toilet that meets or exceeds the 250 g performance threshold should meet customer expectations for flushing.

Before the test, the toilet fixtures were assembled, placed on the test stand, and connected to a water supply (at 50 psi static pressure). Tank water levels were set to the waterline and flush volumes were recorded. Adjustments were made, if necessary, to ensure that all toilet samples flushed with the rated vol-



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ume by adjusting the water level in the toilets. Each toilet was then loaded with test media in 50-g increments until it failed to pass 100% of the media, as seen by the human eye, in two out of three attempts. Four loosely crumpled balls of single-ply toilet paper (six sheets each) were included in each test.

The water change-out rate for all toilets was measured under liquid-only conditions by adding a brine mixture to the water in the bowl, measuring the conductivity of the water, flushing and

remeasuring the conductivity. The difference in conductivity was used to calculate the percentage of water changed out during the flush.

We tested two samples of each toilet model and averaged the results to rate the model's performance.

The testing identified models that clear less than 250 g of media, models that clear 250 g–500 g, and models that clear more than 500 g. Testing was done on gravity flush and 1.6-gallon and 1.1-gallon (6-liter and 4-liter) pressure-assisted toilet models. Mansfield pressure-assisted toilets, as distinct from the other pressure-assisted models, are dual flush; the higher volume mode is to clear solid waste and the lower volume mode is to clear liquid waste. These toilets were tested at both the 1.6-gallon and the 1.1-gallon flush volumes. The test results at 1.1 gallons were compared with the results for other 1.1-gallon pressure-assisted toilets. Test results sorted by performance in clearing solid waste are presented in Table 1.

The test program revealed a significant range in the maximum performance levels of the toilets—yet all of these toilets are certified as meeting the minimum standards set by CSA and ANSI/ASME. All of these toilets met the water exchange requirements of the national standards—indicating that this test may not be meaningful in determining overall effectiveness of flush performance.

It is interesting to note that there was some variability in the performance of the two samples of the same

Table 1. Toilet Performance

	Gravity flush		Dual-flush		1.1-g (4-l) Pressure-assist
	Vacuum-assist		1.6-g (6-l) Pressure-assist		

Make	Model	Model Number	Map Performance
Mansfield	Quantum	148-123 (tank, bowl)	925
Gerber	Ultra Flush	21-302: 21-342 bowl, 28-380 tank	900
TOTO	Drake	CST744S	900
Foremost	Premier	LL-8207-HC bowl, T-8207 tank	850
Mansfield	Quantum	146-123 (tank, bowl)	850
Vitra	Corina	5069-003-0075 bowl, 5070-003-0122 tank	850
Mansfield	Quantum	147-123 (tank, bowl)	825
Vitra	Corina	5068-003-0075 bowl, 5070-003-0122 tank	825
Am. Std.	Glenwall	3402016.021 bowl, 4097100.021 tank	800
Vitra	Altantis	5050 bowl, 6853 tank	800
Am. Std.	Colony Afton	4392.500.020 tank & 3038.016.020 bowl	775
Am. Std.	Cadet	4098.100.020 tank & 3099.016.020 bowl	750
Crane	Economiser	3612 tank & 3824 bowl	750
Mansfield	EcoQuantum @ 1 gal	147 bowl, 119 tank	750
Vitra	Altantis	5051 bowl, 6853 tank	750
Crane	Economiser	3825 bowl, 3612 tank	725
Crane	VIP Flush	3995: 3990 bowl, 3976 tank	725
Niagara	Flapperless	N2216	725
Briggs	Hathaway Vacuity	4360-130 bowl, 4460-130 tank	700
TOTO	Ultramax	MS854114S	700
TOTO	Clayton (nee Baldwin)	ST784S tank & C784SF bowl	675
TOTO	Plymouth	MS924154F	675
Kohler	Cimarron	4287-0 bowl, 4634-0 tank (larger value @ 1.6G, smaller value @ 1.4G)	675
Kohler	Cimarron	4286-0 bowl, 4634-0 tank (larger value @ 1.6G, smaller value @ 1.4G)	450
Briggs	Vacuity	4345-130 bowl, 4400-130 tank	650
Mancesa	Cyclone	2282W bowl	650
Mansfield	EcoQuantum @ 1 gal	148 bowl, 119 tank	650
TOTO	Carusoe	ST706 tank & C715 bowl	650
TOTO	Dalton	ST733 tank & C734F bowl	650
Gerber	Ultra Flush	28-380 tank & 21-374 bowl	625
TOTO	Carlyle	MS874114SG	625
Mansfield	EcoQuantum @ 1 gal	146 bowl, 119 tank	600
Am. Std.	Champion	3121.016.020 Bowl, 4260.016.020 tank	575
Caroma	Tasman	270	550
Crane	Flormont	3992 bowl, 3976 tank	550
Eljer	Aquasaver	1417-00000/137-7025-00	550
Glacier Bay	Westminster	455-685 tank (lined) and 445-684 bowl	550
TOTO	CST703	CST703	550
Vortens	Vienna II	3412 tank, 3207 bowl	550
Briggs	Vacuity	4400-130 tank & 4310-130 bowl	500
Caroma	Caravelle	2000 tank & 270 bowl	500
Crane	SureFlush	31212 bowl, 31242 tank	500
Crane	SureFlush	31202 bowl, 31242 tank	500
Eljer	Patriot Savoy	131-2120-00 bowl, 141-0223-00 tank	500

toilet model. In one-third of the cases, both samples tested gave the exact same result. In one-third of the cases, the results varied by 50 g. And in one-third of the cases, results varied by 100 g between the two samples of the same model.

Testing Flapper Effectiveness

Although toilets can last for more than 20 years, flappers or flush valves may need replacing after 5 years. Many flappers sold for after-market replacement have adjustable closure times and thus the toilets offer adjustable flush volumes. This can be a concern for water agencies that promote low-flush toilets. Some flappers use an adjustable dial, while others use various inserts to adjust the closure time to suit a particular toilet. Although this may make it possible to install the after-market flapper in almost any toilet, the consumer must set the dial correctly, or apply the correct insert in order to achieve the best performance. As a result, it is likely that many low-flush toilets with after-market adjustable flappers are not flushing at 1.6 gallons (6 liters).

Testing flapper performance can give consumers and water agencies that conduct in-residence customer service audits the information they need to adjust replacement flappers properly. In many cases it may not be easy for homeowners to purchase the correct replacement flapper for their toilet fixture. This is particularly true if they are purchasing the flapper at a big-box retail outlet or hardware store, or if they have forgotten the model of their toilet. Installing an incorrect flapper can change the toilet's flush volume, which will affect flush performance. If the toilet flushes with less water after the flapper is replaced, its ability to clear waste may be compromised. If the toilet flushes with more water, its water efficiency may be partly or totally lost.

The MaP project tested three different types of adjustable flapper in each toilet model to determine the appropriate setting (dial or insert) to maintain the design flush volume. Approximately

25 samples of each flapper type were tested (each sample was used for only a few tests). Fluidmaster and Niagara Conservation provided samples of their adjustable flappers, while the Frugal Flush adjustable flapper was available in the Veritec inventory. These are some of the more popular adjustable flappers in the marketplace.

For various reasons, replacement flappers could not be installed in every toilet model. Some models use proprietary flappers, such as the 3-inch (75-millimeter) flappers used in the Toto Drake, Ultramax, and Ultimate; the 2-inch (50-mm) disks used in the Mansfield Alto; and the proprietary flush valve seal used in the American Standard Champion. Other models excluded from the test were pressure-assisted toilets and toilets where existing trim components interfered with the proper operation of the replacement flapper.

To simulate the effects of replacing the original equipment manufacturer (OEM) flapper (often an early-closing flapper) with the commonly available standard or universal flapper, all toilets were fitted, where possible, with a standard after-market replacement flapper. This resulted in a range of flush volumes (see Table 2). The test results show that in more than 70% of the models we tested, flush volume increased by 20%—0.33 gallons—or more when a standard replacement flapper was installed—some models flushed with more than 4 gallons (15 liters). (Yes, believe it or not, some tanks hold more than 4 gallons of water!)

These data indicate that a sizable reduction in water savings will occur over time if homeowners replace early-closing flappers with standard replacement flappers. There is currently some discussion among water efficiency promoters as to whether toilets with early-closing flappers should even be promoted or subsidized by water agencies.

Table 1. Toilet Performance (continued)

Gravity flush	Vacuum-assist	Dual-flush	1.1-g (4-l) Pressure-assist
Make	Model	Model Number	Map Performance
Kohler Am. Std.	Santa Rosa Champion	3323-0 3225.016.020 bowl & 4260.016.020 tank	500 500
Am. Std.	Champion	3110.016.020 bowl & 4260.016.020 tank	475
Vitra	Dual Flush	5076 bowl, 5055 tank	475
Crane	Canada	3352 bowl, 3742 tank	450
Vortens	Genova	3421-02-V tank & 3121-02-V bowl	450
Eljer	Patriot	141-0220-00 tank & 131-2175-00 bowl	425
Eljer	Patriot Savoy	131-2120-00 bowl, 141-0220-00 tank	425
TOTO	Ultimate	MS853113	400
Briggs	Vacuity	4200	375
Crane	Canada	3372 bowl, 3742 tank	375
Foremost	Premier	T-8207-W tank & LL-8207-W bowl	375
Vortens	Dual Flush	3208-02-V bowl & 3420-02-V tank	375
Western Pottery	Aris	822	375
Crane	SureFlush	31192 bowl, 31242 tank	350
Crane	VIP Flush	3976 tank & 3991 bowl	350
Foremost	Regent	T-5207-W tank & LL-5207-W bowl	350
Mancesa	St. Michelle	4260 tank, 2360 bowl	350
Am. Std.	Sonoma	4392.562.020 tank & 3338.012.020 bowl	325
Eljer	Savoy	141-0260-82 tank & 131-2120-82 bowl	325
Kohler	Sterling Rockton	402021-0 bowl, 402022-0 tank	325
TOTO	Ultimate	MS854114	325
Vortens	GTA	3412 tank, 3200 bowl	300
Western Pottery	Clinton	832 bowl	300
Foremost	Victorian	T-1207-W tank & LL-1207-W bowl	275
Mansfield	Alto	130-160	275
Kohler	Devonshire	4269-0 bowl & 4619-0 tank	250
Kohler	Wellworth	K3423 toilet, with 4620 tank/4277 bowl	250
Am. Std.	Plebe	4392.312 tank & 3344.312 bowl	225
Mancesa	Windsor	2700W bowl & 2711W tank	225
Am. Std.	Ravenna	4096.516.020 tank & 3454.016 bowl	200
Corona	Orchid	8510	200
Mancesa	San Marino	2262W bowl & 2261W tank	200
St. Thomas	Marathon	6201.01	200
Glacier Bay	Aragon IV	164963	175
Am. Std.	Cadet	2798.012, with 4112.016 tank & 3454.016 bowl	150
Am. Std.	Hamilton	2092-0170-20	150
Briggs	Abingdon III	4229 = 4440t/4875b	150
Briggs	Altima III	4232 = 4430t/4320b	150
Crane	Cranada	3503 tank & 3415 bowl	150
Crane	Cranada II	3742/3743 tank & 3503 bowl	150
Eljer	Canterbury	081-1620-00	150
Eljer	Patriot	091-2120, with 141-2120 tank/131-2120 bowl	150
Gerber	Aquasaver	21-712, with 28-790 tank	150
Niagara	Niagara Turbo	N2220	150
Orion	Iris	51073 tank & 50073 bowl	150
Am. Std.	Cadet	2898.012, with 4112.016 tank & 3459.016 bowl	125
Kohler	Memoirs	4257-0 bowl & 4454-U-0 tank	125
Kohler	Wellworth	K3422 toilet, with 4620 tank/4276 bowl	125
Komet	Deco	DE 611 tank and DE 627 bowl	75

Table 2. Flush Volume with Standard Flapper Replacements

Manufacturer	Model	Flush Volume	
		Liters	Gallons
American Standard	Hamilton EL	4.9	1.29
Briggs (Proflo)	Vacuity EL	5.3	1.40
Niagara	Turbo RF	5.7	1.51
Crane	VIP Flush RF	6.0	1.59
Crane	Cranada II RF	6.1	1.61
Foremost	Regent RF	6.4	1.69
Briggs (Proflo)	Abingdon III RF	6.8	1.80
Briggs (Proflo)	Altima III RF	6.9	1.82
Foremost	Premier RF	7.3	1.93
Toto	CST703 RF	7.5	1.98
Mancesa	Charleston RF	8.0	2.11
Mancesa	Ste. Michelle RF	8.6	2.27
Glacier Bay	Westminster RF	9.5	2.51
American Standard	Sonoma RF	9.7	2.56
Eljer	Patriot RF	9.9	2.62
Kohler	Wellworth RF	9.9	2.62
Orion	Iris RF	9.9	2.62
Kohler	Wellworth EL	10.0	2.64
Crane	Cranada RF	11.0	2.91
Vortens	GTA RF	12.0	3.17
St. Thomas	Marathon RF	12.2	3.22
American Standard	Plebe EL	12.4	3.28
Western Pottery	Aris RF	12.7	3.36
Sanitarios Azteca (Lamosa)	Sahara RF	13.8	3.65
Gerber	Aquasaver EL	14.0	3.70
American Standard	Cadet RF	15.2	4.02
American Standard	Cadet EL	15.9	4.20

Implications for Builders, Consumers, and Water Providers

These results provide only a snapshot of the effectiveness of low-flush toilets currently available. It would be valuable to bear in mind that

- the models selected do not represent all models currently on the market;
- since only two examples of each model were tested, the results may not represent the performance of that model overall; and
- some models may have been modified by the manufacturer since the testing was completed.

However, two clear conclusions can be drawn from this study. First, the results show that while many low-flush toilets perform well at 1.6 gallons and will meet or exceed customer expectations, there are still too many models certified by standards bodies in the United States and Canada that do not. If consumers and water agencies alike are to feel confident that it is worth

investing in low-flush toilets it may be necessary to raise the requirements for certification.

Since the liquid-exchange test does not appear to indicate how well a toilet performs, certification should not depend heavily on this test. In contrast, the solid-waste test developed for this study is a good indicator of toilet performance and might well be included in certification requirements. Following the methodology used in this study, soybean paste would be used to represent solid waste, and 250 g would be adopted as the threshold for adequate performance.

Second, the study confirmed that many low-flush toilets use more water than the manufacturer claims when set up according to the manufacturer's directions; and that replacing the flapper is likely to affect the flush volume, usually increasing it well above 1.6 gallons. Builders and consumers should be encouraged to purchase models that can be easily set up to flush the appro-

priate volume. They should also consider models that are designed with a standard flapper that can be replaced with an equivalent model from a big-box store such as Home Depot. If they choose a model with a proprietary flapper, consumers should make sure that the proper replacement flapper is readily available. If consumers install an adjustable flapper, it may be difficult for them to identify the proper setting to maintain water savings and flush performance.

The desired water savings are more likely to be realized in water utility low-flush toilet rebate programs when toilets can be easily set up to the correct volume; are equipped with a standard or proprietary flapper, not an adjustable flapper; and perform adequately at the correct flush volume.

Overall, the MaP testing protocol appears to be well received by water providers and manufacturers alike. We expect that many agencies and municipalities will consider the results of MaP testing when evaluating which toilet models to subsidize or rebate. So far, the cities of Toronto and Calgary, in Canada, and Austin, Texas, and Seattle, Washington, in the United States, have used MaP's test results to evaluate toilets, and the Canadian regions of Waterloo, Peel, and Durham are considering it.

Bill Gauley, P.E. is the principal at Veritec Consulting, Incorporated, in Mississauga, Ontario, Canada.

FOR MORE INFORMATION:

Bill Gauley
Veritec Consulting Incorporated
1495 Bonhill Rd., No. 12
Mississauga, ON
Canada L5T 1M2
E-mail: bill@veritec.ca